AUTOMATED DETECTION OF SATELLITE TRAILS IN GROUND-BASED OBSERVATIONS

U-Net segmentation with probabilistic Hough post-processing

Baron Gracias | MPhil Data Intensive Science | University of Cambridge | Supervisor: Dr Eduardo Gonzalez-Solares

Motivation & Scientific Justification

Low-Earth-orbit satellite constellations increasingly cross wide-field astronomical exposures, leaving bright linear trails that corrupt photometry, astrometry, and transient-source catalogues. Manual inspection does not scale to survey operation, so observatories need automated trail masks that are reliable at the pixel level and cheap enough to run inside reduction pipelines.

This project replicates the satellite-trail detector of Stoppa et al. (2024): a U-Net trained on labelled MeerLICHT image patches, followed by a probabilistic Hough transform that reconnects linear trail evidence missed by the network. The replication is used as a diagnostic tool: the aim is not only to reproduce a headline score, but to identify where the residual error comes from.

Methodology

The study uses 178 labelled MeerLICHT image/mask pairs split at image level into 122 training, 24 validation, and 32 test images, avoiding same-image leakage. Images are tiled into 528×528 patches. The U-Net is reimplemented in PyTorch from the released Keras model configuration, preserving average pooling, LeakyReLU activations, in-block dropout, Keras-style initialisation, and a combined binary cross-entropy plus squared-Dice loss.

Hyperparameters are selected by validation-only Optuna search, the top trials are retrained across five seeds, and the final threshold is chosen from a validation precision–recall sweep before test metrics are inspected. Hough post-processing is evaluated on a reconstructed patch canvas, so its numbers are reported as patch-canvas approximations to dense full-image inference with separate precision and completeness caveats.

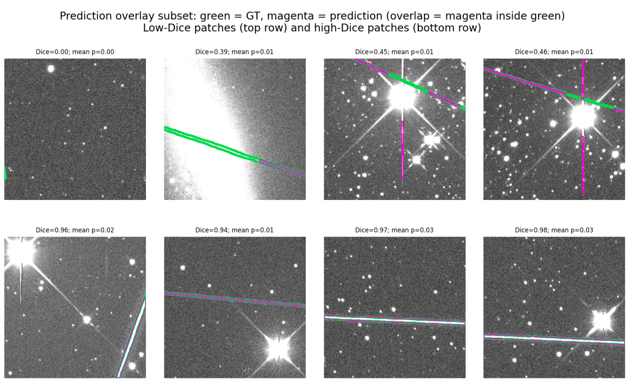

Figure 1: Representative test-patch predictions for the locked U-Net: ground truth in green, prediction in magenta, and overlap inside the green band.

Key Findings

The replication reproduces the target paper's recall closely: five-seed sampled-test recall is 0.923 ± 0.006 against the paper's reported 0.94. Strict precision is lower, at 0.847 ± 0.009 on the sampled test set and 0.780 on the parity set that restores the natural negative-patch prevalence. This shows that precision is prevalence-sensitive, whereas recall is the cleaner like-for-like comparison across the two test populations.

The Hough stage reproduces the paper's intended completeness behaviour. On the canonical patch canvas, trail-pixel recall rises from 0.918 to 0.990, equivalently reducing the pixel-level false-negative rate from 0.082 to 0.0105. This does not make the Hough mask a better strict pixel classifier: post-Hough line drawing increases false-positive area, reducing strict precision from 0.780 to 0.410 on the parity canvas.

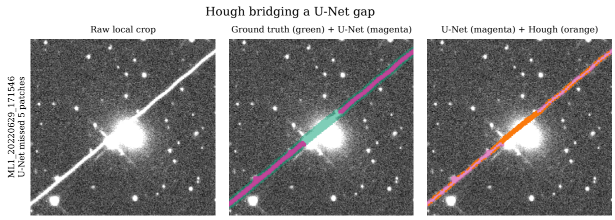

Figure 2: Hough bridging a U-Net gap over a bright star. The example illustrates the mechanism: linear post-processing recovers ground-truth trail pixels missed by the thresholded U-Net.

The strict precision gap is localised to boundary scale. False-positive decomposition shows that only 17.5 % of false-positive pixels occur in trail-free patches; the remaining 82.5 % are over-painting within trail-bearing patches. A one-pixel boundary tolerance raises five-seed precision from 0.847 to 0.946, and 57 % of all false positives lie within one pixel of a labelled trail. These diagnostics are consistent with boundary placement and mask granularity dominating the residual error, although a gold-standard re-annotation audit is needed to prove that interpretation.

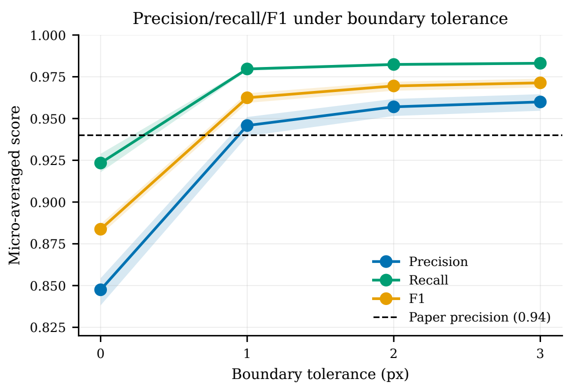

Figure 3: A one-pixel tolerance largely closes the strict precision gap, indicating that the residual error is concentrated around thin mask boundaries.

Extensions

Follow-up experiments test whether the residual can be moved by model or inference changes. A classifier-gated detector shifts precision by only +1.2 percentage points, consistent with the 17.5 % inter-patch upper bound. An attention-gated U-Net at matched capacity is a null result, clDice shows no hidden topology gain, hard-negative mining adds only +0.007 precision, and ensembling with test-time augmentation sits within seed noise.

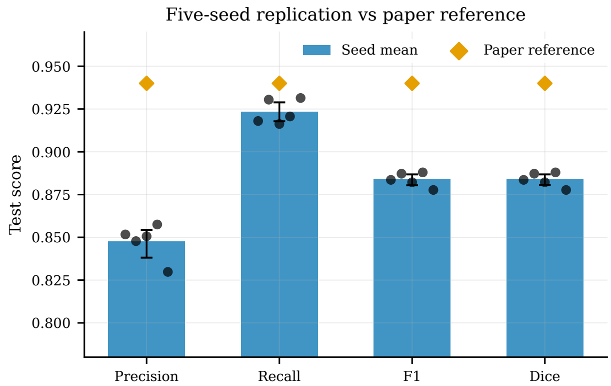

Figure 4: Five-seed replication performance for the locked U-Net compared with the Stoppa et al. reference. Recall is closely reproduced, while strict precision remains lower, motivating the boundary-scale error analysis.

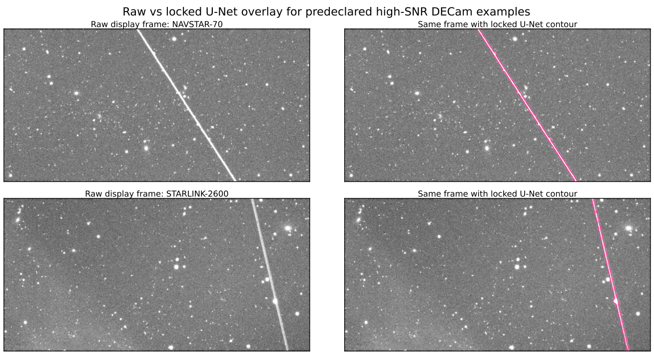

Figure 5: Cold-domain DECam examples: raw measured-streak frames and the locked MeerLICHT-trained U-Net contour. This is a qualitative transfer illustration, not a benchmark.

Recommendation & Next Steps

The decisive next experiment is a gold-standard re-annotation audit of a small test subset, with explicit mask-width conventions. That would separate true detector error from label-boundary convention. A labelled cross-instrument test, especially on DECam-like data, is the next generalisation check; the current DECam application is qualitative only.

A threshold-0.58 row is useful only as a diagnostic symmetry check, not a new selection result. A resolution experiment is feasible as a synthetic resampling stress test, but should not be presented as cross-instrument evidence without new labels.

Research Impact & Conclusion

The U-Net plus Hough pipeline faithfully recovers the target detector's recall and Hough completeness. The tested capacity, training-protocol and post-processing changes do not materially reduce the remaining strict-precision gap; it is bounded and concentrated within about one pixel of the labelled boundary. For operational masking, this means reporting strict scores alongside boundary-tolerant scores and annotation conventions is more informative than another architecture tweak.

References

[1] F. Stoppa et al., "Automated detection of satellite trails in ground-based observations using U-Net and Hough transform," *Astronomy & Astrophysics*, 692, A199, 2024. [2] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," MICCAI, 2015. [3] J. Matas, C. Galambos, and J. Kittler, "Robust detection of lines using the progressive probabilistic Hough transform," *Computer Vision and Image Understanding*, 78(1), 2000.

Declaration of Use of Autogeneration Tools

Codex (ChatGPT) and Claude Code supported library usage, plotting and scripting boilerplate, function docstrings, and project-structure review. All prose and reported numbers were manually reviewed, and all metrics derive from the repository artefacts.